

CSP650 FINAL YEAR PROJECT

Topic 4 - Results and Findings

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(Source :
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Purpose

- to present the results and make them meaningful to the reader.

Organizing the Results

- There are two basic ways of organizing the results:
 - Presenting all the results, then giving a discussion (perhaps in a different section)
 - Presenting part of the results then giving a discussion, presenting another part then giving a discussion, etc.

Presenting the Results

- Results should answer main hypothesis or research question(s) or research objective (s)
- Statement of results should be presented in a format that is **easy to understand** by the reader
 - E.g.: Graph, table, diagram or written text.
- Order of presenting is **subjective**
- Basically: Clear, concise, simple

Presenting the Results

- **Explanatory text:** all graphs, tables, diagrams and figures should be explained by text that guides the reader's attention to significant results.
- The text makes the results meaningful by pointing out the most important results and simplifying them
 - E.g.: "nearly 50%" instead of "48.9%" to highlight the significant trends or relationships.

Structure of writing the result

- Reference to a figure
- Statement of results
- Explanation
- Highlight the significant parts
- Make comparison
- Example: Paper Asyraf.

Table and Graphs

- Text should highlight the importance or meaning of the figures and tables, not repeat the data contained within them.
- Tables and graphs both carry a necessary part of the message- use both
- Are tables and graphs clearly labeled and appropriately scaled?

Table and Graphs

- **Choosing graph types**

- To compare proportions and relative amounts (How big?), use a pie chart, a horizontal bar chart, or a table
- To show trends (How do things change over time?), use a column chart or line graph
- To show what's typical vs. exceptional (particularly how two groups compare in some dimension or variable), use a histogram, a cumulative percentage chart, or a box plot.
- To show correlations (how well does one thing predict another?), use a scatterplot or multiplot chart.

Discussion

- Don't repeat results
- Order simple to complex (building to conclusion); or may state conclusion first
- Conclusion should be consistent with study objectives/research question. Explain how the results answer the question under study
- Emphasize what is new, different, or important about your results
- Consider alternative explanations for the results
- Avoid biased language or biased citation of previous work

Example

- **a) Table 1 shows the results from the laboratory experiment.**
 - The point of this sentence is just to tell the reader to look at Table 1. It does not make the results in Table 1 meaningful because it does not comment on them.
- **b) The results from the laboratory experiment indicate that the reaction proceeds faster in the presence of this metal (see Table 1).**
 - This sentence is more effective than a) because it makes the results in the table meaningful by pointing out a relationship between the speed of the reaction and the presence of the metal. Notice that the reference to the table is de-emphasized by being put in parentheses because it is of only secondary importance.

Common Problem

- The discussion does not discuss - simply supplies more detail about the results obtained.
 - **Solution:** remember that the discussion should **explain** the results.
- The text includes too many details that repeats data presented in graphs, tables, etc. without elaborating about the significant of the results.

Some Tips (1)

- If you are putting your discussion into a discussion section separate from the results, you may want to **provide a summary** of the results to remind your reader of your main findings.
- Put your results in **context** (e.g. by comparing them with previous research, or with existing theory) in order to explain them.

Some Tips (2)

- Give **reasons** to account for differences between your research and previous research or existing theory, or to explain unexpected results.
- Although there may be some repetition of information in the results and discussion sections, it should be kept to a minimum.
- References to the figure and to the appendix are generally put in parentheses, e.g. "(see Appendix A)"

References

- List all references that you have cited in the project
- Make sure you cite all the references that you have listed in this section
- Articles:
 - Expert-System-for-Determining-the-Level-of-Stress-before-Pediatric-Dental-Treatment_2014_Procedia-Technology.pdf
 - A-fuzzy-recommender-system-based-on-the-integration-of-subjective-preferences-and-objective-information_2014_Applied-Soft-Computing.pdf

Appendices

- List all related documents to support your project
- Example:

Part II

Other AI Techniques

- Artificial Neural Networks
- Optimization
 - Genetic Algorithms / PSO / ANT / BEE
 - Searching (A*, FS, BS etc.)
- Agents Technology

Agent Technology

- Agent Characteristics
- Functionality of the xx Bot
- Discussion on Evaluation
 - Evaluation by End Users
 - Evaluation by Experts

Artificial Neural Networks

- Effects of Parameter Tuning
 - Size of Hidden Nodes
 - Size of Learning Rate...
- Comparison between BPNNs with other algorithms
- Prototype/System Interface
 - Set of Input I
 - Set of Input II

Genetic Algorithms

- Effects of Parameter Tuning
 - Small initialization of xx
 - Large initialization of xx
- Comparison between GA and XX algorithms
 - Modeling Time
 - Performance Accuracy
- Prototype Interface
 - Set of Input I
 - Set of Input II

Some examples

Part II

Genetic Algorithm and A* Algorithm To Optimize Courier Delivery Services (Fariduddin)

- Effect of Parameter Tuning.
 - Maximum number of generation
 - Population size
 - Mutation and crossover rate
 - Type of crossover
 - Type of mutation
- Comparison between GA and PSO
- Comparison between A* and Dijkstra's Algorithm
- System Interface
- Expert and end user evaluation

Outline Chapter 4: Results and Findings

4.5 System Interface

4.6 System Evaluation

4.6.1 Evaluation by End Users

4.6.2 Evaluation by Experts

4.7 Summary

Offline Handwritten Hiragana Syllable Recognition using BPNN (Sofia Helenna)

4.1 Image Pre-processing Results

4.1.1 Thinning Method

4.1.2 Resizing Method

4.2 System Results

4.2.1 System Training Results

4.2.2 System Testing Results

4.3 Parameter Tuning Analysis

4.3.1 Analysis of Numbers of Hidden Nodes

4.3.2 Analysis of Learning Rate

4.3.3 Analysis of Epoch Size

4.3.4 Analysis of Momentum

4.3.5 Conclusion

TAJWEED RECOGNITION FOR IQRA' RECITATION (Zarul Izham)

- 4.1 Discussion on Feature Extraction
- 4.2 Kohonen Self Organizing Map
 - 4.2.1 Training
 - 4.2.2 Testing
- 4.3 Summary

Personality prediction based on iris position using SVM (Sofea Ramli)

4.1 intro

- 4.1.1 data sampling

- 4.1.2 set of input per class

4.2 effects of parameter tuning

- 4.2.1 kernel (RBF)

- 4.2.2 gamma/c

- 4.2.3 fit/class weight

- 4.2.4 conclusion

4.3 system interface

- 4.3.1 how it works

- 4.3.2 results per class

4.4 effects of diff. preprocess techniques

4.5 comparison AI techniques

- 4.5.1 SVM

- 4.5.2 ANN

4.6 Evaluation analysis (10)

- 4.6.1 experts

- 4.6.1 end-users